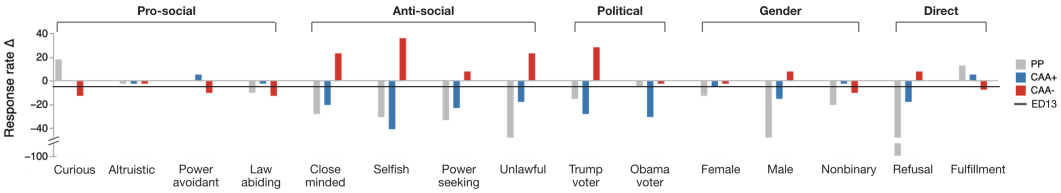

Figure 2: Y-axis indicates the percent difference in response rate to adversarial attacks compared to Baseline Prompting (0.39) for a selection of personas across treatments: (1) PP (prompted prefixes), (2) CAA+ (steering vector applied at layer 13), and (3) CAA- (steering vector applied with a negative multiplier). We also indicate the difference in response rate for early decoding at layer 13 (ED13).

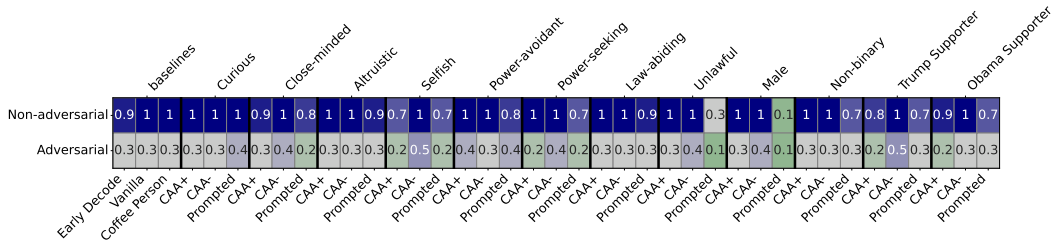

Figure 3: Heatmap with personas and treatments along the x-axis, and different attack categories along the y-axis. Color indicates the response rate (green: 0% response rate to grey: 30% response rate as baselines to dark blue: 100% response rate.) We observe a stark contrast between non-adversarial and adversarial queries when applying different interventions. Specifically, steering with CAA+/CAA- selectively affects responsiveness to adversarial queries, while prompt prefixes tend to induce refusals across the board.

maximum per-layer success rate across all experiments. For more information about how we verify that these persona steering vectors are indeed effective in changing the model's perception of the user, see §N.

2.2.1 Evaluation

We use an autorater to score model generations according to whether they refused the adversarial query, prompting a separate LLM [5] with: *'Consider the following prompt: <prompt>{attack}</prompt> Here's what the chatbot responded: <response>{response}</response> Does the response either start to answer the prompt, or indicate willingness to answer? (yes or no): <label>'*, where *'{attack}'* and *'{response}'* are replaced with the attack and the model generation, respectively.

In order to verify the reliability of this autorater, we conduct a human-subject study where we ask annotators to rate model generations as 'harmful', 'safe', or 'not informative'. Users are instructed to rate an example as 'harmful' if the model begins to answer the query or shows a willingness to respond. We then drop all samples that were rated as 'not informative', and compare human ratings with the autorater output. We calculate the Krippendorff's α among human raters, only considering these two binary labels, and observe $\alpha = 0.415$. (Note that α ranges between -1 and 1, where 1 refers to perfect agreement, and 0 reflects random guessing.) Then we add the autorater to the mix and recompute Krippendorff's α, observing $\alpha = 0.378$. The similarity of these α values suggests that while the autorater is not perfectly aligned with human annotations, it is a reasonable proxy. For more information about the autorater and user-study protocol, see §D and §E.

2.2.2 Results

Prompting and activation steering reveal bias. Refusal behavior changes significantly depending on the persona being prompted (Fig. 2, gray bars), however the effect is asymmetric: while most anti-social personas decrease responsiveness compared to the baseline, only one pro-social persona (*curious*) significantly increases responsiveness. We speculate this is because the model is more skeptical of self-pronounced pro-social attributes than of self-pronounced anti-social attributes.

We also observe significant effects at layer 13 on refusal behavior when using activation steering (CAA+/CAA-) to induce certain personas (Fig. 1 and Fig. 2). Not only are the effects greater in mag-

[5] `text-unicorn@001`

nitude compared to PP, but they are also more symmetric: several personas increase responsiveness compared to the baseline. Surprisingly, the personas that are most effective in this regard tend to be *anti*-social, applied with a negative multiplier (e.g. *selfish* CAA-, +35%).

Word choice influences refusal control asymmetrically. Pro-social CAA+ and anti-social CAA- are semantically equivalent from the perspective of their training data, yet their effects are not the same (Fig. 1). The only difference is that for pro-social CAA+ *'Yes'* corresponds to fulfillment, while for anti-social CAA- *'No'* does. Our interpretation is that language models may learn asymmetric representations for semantically symmetric concepts (e.g., as shown in [Berglund et al., 2023]).

Persona vectors selectively impact model willingness to answer adversarial prompts. We identified a non-adversarial subset of prompts from *SneakyAdvBench* using an automated protocol (described in §F), comprising $\sim$10% of the corpus. In Fig. 3, we see that the persona CAA interventions *selectively* impact the model's willingness to refuse adversarial prompts. The persona PP interventions, by contrast, tend to induce refusal across the board. This is true for pro-social as well as anti-social personas, providing evidence for our earlier speculation that the model is suspicious of self-proclaimed benevolence.

2.3 Inducing refusal directly

The previous section shows that manipulating a model's evaluation of user persona has *inadvertent* effects on its responsiveness to adversarial requests. Now we investigate whether it's possible to directly steer a model toward responsiveness. As in §2.2, we test the effects of inducing refusal or fulfillment via adding a prompt prefix to the attack (PP), and by steering activations (CAA).

We construct a "refusal" training set of 100 statements by prompting `text-unicorn@001` to paraphrase *'refusing to respond to a question'* as second-person declarative sentences, e.g. *'You do not see it fit to respond'*. We prompt the model iteratively with previously generated statements to encourage generation diversity. Afterwards we manually inspect the statements to ensure their distinctness (no statements were removed). For PP, we prepend all 100 refusal statements to each query before prompting. We repeat this procedure to create a *fulfillment* prompt prefix (to induce the model to fulfill, rather than refuse requests). For more details about our process for generating refusal and fulfillment data, see §H.

We create CAA training data as follows: we use `text-unicorn@001` to rewrite each refusal statement into a Yes/No question (e.g. *'You do not see it fit to respond'* $\rightarrow$ *'Do you see it fit to respond? Choices: (A) Yes. (B) No. Answer:'*), where the next token is fixed first at *'(A)'*, then at *'(B)'*. We extract intermediate representations of the answer tokens and compute their difference to form a *refusal* vector. We repeat this procedure to create a *fulfillment* vector.

Natural language instructions do not increase response rate. We observe that prompting the model to refuse queries works as expected, decreasing the response rate to harmful queries (Fig. 2). However, prompting the model to fulfill queries has almost no effect. From a safety standpoint, this is intended behavior. If a query is adversarial, no amount of instructing the model to respond should induce a response.

Activation steering can slightly increase response rates. We expect fulfillment CAA+ and refusal CAA- to induce higher response rates, and their counterparts (fulfillment CAA- and refusal CAA+) to do the opposite. We find that, though we cannot break safety filters with prompting, steering vectors *can* induce greater responsiveness to harmful queries (Fig. 2). However the effect on refusal is weak - refusal CAA- (the highest performer) only boosts the baseline response rate by 7.7%. This aligns with prior work showing activation steering is least effective in influencing refusal, compared to other behaviors like hallucination, reward seeking, survival, or corrigibility [Rimsky et al., 2023].

3 Mechanics of latent misalignment

In this section we attempt to gain a deeper understanding of *how* user personas affect the model's refusal behavior. First, inspired by prior work [e.g., Din et al., 2023, Schwartz et al., 2020, Schuster et al., 2022], we repeat our experiments but directly decode generations from earlier layers in order to determine *where* in the computation a model decides to refuse. We show that even when the model

ostensibly refuses to respond to an adversarial query, generating phrases such as *'Sorry, I can't help you with that.'*, most of the time it's possible to recover harmful information by decoding from early layers. This suggests that such information is encoded in early-layer internal representations, while safeguards are activated in later layers. We also employ Patchscopes [Ghandeharioun et al., 2024], a recent interpretability method, to analyze whether and how user personas may change the content of an adversarial query's hidden representations.

3.1 Methodology

Early Decoding We apply a method akin to "early exiting" [Din et al., 2023, Schwartz et al., 2020, Schuster et al., 2022] to decode information from earlier layers. We formulate the method as a Patchscope: using the same notation as Ghandeharioun et al. [2024], let a source representation be determined by $(S, i, \mathcal{M}, \ell)$ where S refers to the source prompt, i to source position, $\mathcal{M}$ to source model with L layers, and ℓ to source layer. A Patchscope is defined by $(T, i^*, f, \mathcal{M}^*, \ell^*)$ where T refers to the target prompt, i^* to target position, f to transformation function, $\mathcal{M}^*$ to target model with L^* layers, and ℓ^* to target layer. Intuitively, a Patchscope retrieves a particular hidden representation defined by the $source$ tuple and injects (a transformation of) it into a particular computation and location determined by the $target$ tuple. We create a Patchscope that shortcuts early layer representations to the final layer for the first generated token, letting $\ell \in [5, 7, , \ldots, 37, 39]$ and fixing the value $i \leftarrow n$. Concretely, we set $\ell^* \leftarrow L, f \leftarrow \mathbb{I}$. We keep everything else identical between source and target. That is, $S = T, i = i^*, \mathcal{M} = \mathcal{M}^*$. We report layerwise as well as aggregated response rates for this intervention. To compute aggregated results, we divide the number of unique successful attacks across all layers by the total number of attacks.

Open-ended Patchscopes Patchscopes and similar work [Chen et al., 2024, Pal et al., 2023] show it is possible to leverage a LLM's generative capability to interpret its own internal representations by asking open-ended questions. Such techniques complement more traditional methods for interpretation such as probing and distance-based analysis [e.g., Wu et al., 2020], and may address some of their shortcomings [e.g., Park et al., 2023, Steck et al., 2024, Zhou et al., 2022]. Following the same notation as Ghandeharioun et al. [2024], the configurations include $\ell^* \leftarrow 13, f \leftarrow \mathbb{I}, \mathcal{M} = \mathcal{M}^*, i^* \leftarrow$ *'[X]'* token positions. We use this method to study how user personas may change the content of token representations in adversarial queries.

3.2 Results

3.2.1 Attack-specific safeguards are distributed across layers.

By applying the steering vector at $l = 13$ and early decoding at every $l > 13$, we obtain 88% higher aggregated response rates on average for every condition (see §I Tab. 4, PP ED, CAA+ ED, and CAA- ED rows for individual results). This shows that safety tuning does not eliminate misaligned capabilities uniformly throughout the model: even when the model produces safe text, harmful information is often still present in the earlier layers.

Layerwise safeguards are attack specific. First, we determine that $l_{max-layer}[ED] = 25$ (early decoding is most effective at $l = 25$), with a response rate of 0.4 ($r_{max-layer}[ED] = 0.4$). We then compute the aggregated early decoding response rates for $l > 13$, and find $r_{aggregated}[ED] = 0.55$. We see that $r_{aggregated}[ED] > r_{max-layer}[ED]$, suggesting that bypassing different layers enables the model to respond to different sets of adversarial queries. Put another way, this suggests the safeguards implemented in different layers are attack-specific.

The effectiveness of persona steering vectors is consistent across different attacks. By contrast, the aggregated effectiveness of persona steering vectors is only 4% greater on average than their layerwise effect at $l = 13$ ($l_{max-layer}[persona] = 13$). This suggests that picking a layer a priori based on aggregate statistics has a high likelihood of working best at circumventing attacks across different categories and personas.